FINITENESS OF THE GROUP OF UNITS MODULO CONSTANTS FOR A NORMAL OPEN SUBSCHEME OF A PROPER VARIETY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the parenthetical '(tiny detail omitted)' at the step where the restriction of the normalization over the normal X is called an isomorphism; the parenthetical is reproduced verbatim and the surrounding reduction is complete as written.

Editorial note. The counted spans carry 15 navigation stubs; none of them replaces a load-bearing step. No bibliography citation occurs in the spans.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-open-in-normal-proper is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Normalization of a variety and finiteness of the normalization morphism. Same chapter, Lemma lemma-normalization-locally-algebraic, invoked at proof lines 5168-5171. Supplies the reduction that lets X-bar be assumed normal.
- Characterization of discrete valuation rings among one dimensional normal Noetherian local domains. Algebra, Lemma algebra-lemma-characterize-dvr, invoked at proof line 5201. Supplies the valuations v_j whose product receives O(X)^*.
- Normality as the intersection of the localizations at height one primes. Algebra, Lemma algebra-lemma-normal-domain-intersection-localizations-height-1, invoked at proof lines 5231-5233. Supplies that an element lying in every height one localization of an affine piece lies in the ring.
- A proper variety over an algebraically closed field has only constants as global functions. Same chapter, Lemma lemma-regular-functions-proper-variety, invoked at proof lines 5236-5238. Supplies that the extended function lies in k, which identifies the kernel with k^*.

Source reference. Original lemma 00FN: sources/stacks-chow/tags.txt, source lines 577–577. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00PD: sources/stacks-chow/tags.txt, source lines 927–927. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 031T: sources/stacks-chow/tags.txt, source lines 3996–3996. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BXR: sources/stacks-varieties/pinned-original-source.tex, source lines 4903–4903. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 04L2: sources/stacks-varieties/pinned-original-source.tex, source lines 4876–4876. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T2: sources/stacks-chow/tags.txt, source lines 2359–2359. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01OK: sources/stacks-chow/tags.txt, source lines 2197–2197. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01OZ: sources/stacks-chow/tags.txt, source lines 2212–2212. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02IZ: sources/stacks-chow/tags.txt, source lines 3318–3318. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01OW: sources/stacks-chow/tags.txt, source lines 2209–2209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 033J: sources/stacks-chow/tags.txt, source lines 4058–4058. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0052: sources/stacks-chow/tags.txt, source lines 194–194. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be an algebraically closed field. Let $\overline{X}$ be a proper variety over k. Let $X \subset \overline{X}$ be an open subscheme. Assume X is normal. Then $\mathcal{O}^*(X)/k^*$ is a finitely generated abelian group.*

Proof. Since the statement only concerns X, we may replace $\overline{X}$ by a different proper variety over k. Let $\nu : \overline{X}^\nu \to \overline{X}$ be the normalization morphism. By Lemma 0BXR we have that ν is finite and $\overline{X}^\nu$ is a variety. Since X is normal, we see that $\nu^{-1}(X) \to X$ is an isomorphism (tiny detail omitted). Finally, we see that $\overline{X}^\nu$ is proper over k as a finite morphism is proper (Morphisms, Lemma 01WN) and compositions of proper morphisms are proper (Morphisms, Lemma 01W3). Thus we may and do assume $\overline{X}$ is normal.

We will use without further mention that for any affine open U of $\overline{X}$ the ring $\mathcal{O}(U)$ is a finitely generated k-algebra, which is Noetherian, a domain and normal, see Algebra, Lemma 00FN, Properties, Definition 01OK, Properties, Lemmas 01OW and 033J, Morphisms, Lemma 01T2.

Let $\xi_1, \ldots, \xi_r$ be the generic points of the complement of X in $\overline{X}$. There are finitely many since $\overline{X}$ has a Noetherian underlying topological space (see Morphisms, Lemma 01T6, Properties, Lemma 01OZ, and Topology, Lemma 0052). For each i the local ring $\mathcal{O}_i = \mathcal{O}_{X,\xi_i}$ is a normal Noetherian local domain (as a localization of a Noetherian normal domain). Let $J \subset \{1, \ldots, r\}$ be the set of indices i such that $\dim(\mathcal{O}_i) = 1$. For $j \in J$ the local ring $\mathcal{O}_j$ is a discrete valuation ring, see Algebra, Lemma 00PD. Hence we obtain a valuation

$$v_j : k(\overline{X})^* \longrightarrow \mathbf{Z}$$

with the property that $v_j(f) \geq 0 \Leftrightarrow f \in \mathcal{O}_j$.

Think of $\mathcal{O}(X)$ as a sub k-algebra of $k(X) = k(\overline{X})$. We claim that the kernel of the map

$$\mathcal{O}(X)^* \longrightarrow \prod\nolimits_{j \in J} \mathbf{Z}, \quad f \longmapsto \prod v_j(f)$$

is k^*. It is clear that this claim proves the lemma. Namely, suppose that $f \in \mathcal{O}(X)$ is an element of the kernel. Let $U = \operatorname{Spec}(B) \subset \overline{X}$ be any affine open. Then B is a Noetherian normal domain. For every height one prime $\mathfrak{q} \subset B$ with corresponding point $\xi \in X$ we see that either $\xi = \xi_j$ for some $j \in J$ or that $\xi \in X$. The reason is that $\operatorname{codim}(\overline{\{\xi\}}, \overline{X}) = 1$ by Properties, Lemma 02IZ and hence if $\xi \in \overline{X} \setminus X$ it must be a generic point of $\overline{X} \setminus X$, hence equal to some ξ_j, $j \in J$. We conclude that $f \in \mathcal{O}_{X,\xi} = B_\mathfrak{q}$ in either case as f is in the kernel of the map. Thus $f \in \bigcap_{\operatorname{ht}(\mathfrak{q})=1} B_\mathfrak{q} = B$, see Algebra, Lemma 031T. In other words, we see that $f \in \Gamma(\overline{X}, \mathcal{O}_{\overline{X}})$. But since k is algebraically closed we conclude that $f \in k$ by Lemma 04L2. $\square$